COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CpE21S2
DATE PERFORMED:	2/9/2029	**DATE SUBMITTED:**	2/9/2029
NAME:	Fernandez, Cedric S.	**INSTRUCTOR:**	Engr. Jimlord Quejado

Seatwork 7.2: Algorithm Analysis: Best-Case and Worst-Case Time Complexity

1. PROCEDURE OUTPUT

Instructions: Analyze each algorithm carefully. For every item, determine the following:

1. **Best-Case Time Complexity**
2. **Worst-Case Time Complexity**
3. **Brief explanation of how you arrived at your answers**

Use asymptotic notation such as O(1), O(log n), O(n), O(n log n), O(n^2), etc.

1. Finding an Element

```
C/C++
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }
}
```

Best case: O(1)
Worse case: O(n)
Explanation: In best case the item we need is at the very beginning of the list it will only execute 1 time but for the worse case it will need to check every single item thus n.

---

2. Printing All Elements

```C/C++
for (int i = 0; i < n; i++)
{
    cout << arr[i] << endl;
}
```

Best case: O(n)
Worse case: O(n)
Explanation: There is no condition to end early that is why for the best and worst case time complexity it will be n because it will go on indefinitely.

3. Nested Loops

```None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        cout << i << " " << j << endl;
    }
}
```

n + n = n^2
Best case: O(n^2)
Worse case: O(n^2)
Explanation: For both worst case and best case would depend only on the value of n regardless if there is an for loop inside of a for loop and since it was not given any it shall only run indefinitely until it wont anymore.

4. Conditional Processing

```C/C++
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i];
    }
}
```

```
else
{
    cout << "Small input";
}
```

Best case: O(1)
Worse case: O(n)
Explanation: If our N is less than 100 hundred it will only execute 1 time but if its more then 100 it will print all the items inside arr[i] indefinitely and will only stop based on the i.

5. Fixed Inner Loop

C/C++
```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
}
```

Best case: O(n)
Worse case: O(n)
Explanation: the outer loop runs n times and the inner only runs 10 times so in the end and since for any condition the inner loop will still run 10 times our best case and worst case will depend on the outer loop so n.

6. Two Consecutive Loops

C/C++
```
for (int i = 0; i < n; i++)
{
    cout << arr[i];
}

for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
```

N + n = 2n

Best case: O(n)
Worse case: O(n)
Explanation: both for loops run indefinitely because they have no condition stopping them so we would have n + n = 2n and since we don't care about 2 ib=n cases like this we can drop it leaving only n

7. Dividing by Two

```C/C++
int x = n;

while (x > 1)
{
    x = x / 2;
}
```

1 +
Best case: O(log n)
Worse case: O(log n)
Explanation: For worse and best case it needs to be both decided by x with base 2 making it less than 1

8. Nested Loop With Different Limits

```C/C++
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < i; j++)
    {
        cout << i << j;
    }
}
```

Best case: O(n^2)
Worse case: O(n^2)
Explanation: for both worse case and worse case the inner loop will continue until the loop outside is equal to n making them both run indefinitely so n^2

9. Early Termination

```C/C++
for (int i = 0; i < n; i++)
```

```
{
    if (arr[i] == target)
    {
        cout << "Found";
        break;
    }
}
```

Best case: O(1)
Worse case: O(n)
Explanation: The best case if done if the target found is first in the index making it 1 but if it isn't it will run indefinitely until found so n

10. Two-Level Search

C/C++
```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] == target)
        {
            break;
        }
    }
}
```

Best case: O(1)
Worse case: O(n^2)
Explanation: The best case is O(1) because the loop will break if the target is found at index 1 and the worst case is O(n^2) because the loop will go through 2 for loops for n x n times which is n^2 if the target is at the last.

11. Linear Loop With Logarithmic Inner Work

C/C++
```
for (int i = 0; i < n; i++)
{
    int x = n;

    while (x > 1)
```

```
    {
        x = x / 2;
    }
}
```

Best case: O(N log n)
Worse case: O(N log n)
Explanation: for both worse with and best case the for loop will run indefinitely then it needs by x with base 2 making it less than 1 so n for the first part and log n

12. Conditional Nested Loop

```C/C++
if (arr[0] == target)
{
    cout << "Found";
}
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << arr[i] + arr[j];
        }
    }
}
```

Best case: O(1)
Worse case: O(n^2)
Explanation: for best case the target is found on the first item but for worse case we would need to iterate every single one and since there are two for loops it makes it n^2

13. Finding the Maximum

```C/C++
int max = arr[0];

for (int i = 1; i < n; i++)
```

```
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
}
```

Best case: O(n)
Worse case: O(n)
Explanation: both in worse and best case are n because they all run indefinitely because it still runs regardless of n

14. Three Nested Loops

C/C++
```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        for (int k = 0; k < n; k++)
        {
            cout << i + j + k;
        }
    }
}
```

Best case: O(n^3)
Worse case: O(n^3)
Explanation: because it has 3 for loops since it's both unconditional all of them will run indefinitely.

15. Comprehensive Analysis

C/C++
```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
```

```
            return i;
        }

        int x = n;

        while (x > 1)
        {
            x = x / 2;
        }
    }
```

Best case: O(1)
Worse case: O(n log n)
Explanation: For best case the value can be immediately found in the first item in the index but for the worse case the if loop will continue until found and it will be divided by base 2 until it reaches 1

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.